IMU-Based Camera Stabilisation: Tilt and Pan Control

Lab 2 – open-loop control with a BNO055 IMU and two servos

Embedded Real-Time Control laboratory – M.Sc. Control Systems Engineering, University of Padova
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Purpose

Implement open-loop control on the T-Bot: use an inertial measurement unit to stabilise the tilt of a camera mount and to drive its pan axis, then improve the behaviour with two bonus tasks. The IMU is a BNO055 read over I^2C; the two axes are actuated with PWM servos.

What I implemented

- **Exercise 1 – tilt stabilisation from the accelerometer:** the tilt angle is estimated from the measured Y acceleration, $\theta = \arcsin(a_y/g)$ converted to degrees, and the camera command is the negative of that angle, clamped to $\pm 45°$.
- **Bonus 1 – smooth pan control from the gyroscope:** the pan command is a pure proportional law on the Z-axis angular rate, $u_{pan} = -k_p\,\omega_z$ with $k_p = 30$ (the value I tuned during the lab), also clamped to $\pm 45°$.
- **Actuator mapping:** each angle is mapped to the servo duty as $150 + \theta \cdot (50/45)$ counts and passed through a `saturate()` function that enforces the servo limits, so the mechanical range is never exceeded.
- **Telemetry:** I restructured the datalogger record to carry the interesting signals (`tilt_angle`, `y_accelerate`, `pan_control`, `z_gyro`) and fixed the order of the logger update/send calls so that the transmitted packets are complete; data was acquired with the MATLAB serial/WiFi datalogger for analysis.
- **Bonus 2 – sensor fusion:** combining accelerometer and gyroscope information for a more stable tilt/pan response.

Results

The measured step responses for the exercise and for both bonuses are included as figures in the report: the tilt loop holds the camera level for moderate platform inclination, and the gyro-based pan follows the rotation rate far more smoothly than the accelerometer-only solution. The report documents the control laws, the clamping strategy and the measurement results.

Repository contents

`report/Embedded-Report-lab2.pdf` (submitted report, LaTeX source and figures included) · `src/user_code_main.c` (USER CODE sections of my `main.c`; course framework not redistributed) · `docs/` (this overview).

Skills demonstrated

IMU integration over I^2C (BNO055), inclinometer estimation, proportional control design and gain tuning, accelerometer/gyroscope sensor fusion, PWM servo actuation and saturation handling, fixed-rate real-time loops, measurement and analysis with MATLAB.